

HOW HALLUCINATION DETECTION FAILS ACROSS LLMs

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ABSTRACT

Hallucination detectors for long-form multimodal generation are almost always evaluated on text from the same source LLMs they were trained on. We show this conflates detection with identifying which LLM wrote the text. We release HALT (Hallucination Across LLMs and Techniques), a benchmark of 19,361 forensic video reports (21.8M words) produced by three frontier multimodal LLMs across eight prompting strategies on 807 surveillance videos, labelled for six hallucination types spanning fabrication, omission, and distortion by a cross-judging panel in which no model evaluates its own output. The corpus is fully crossed: every video is described by every source LLM under every strategy, so any cross-source comparison holds the input fixed. Under the standard same-source split, fabrication is the most detectable axis (AUC 0.89) and distortion the least (0.77). Under leave-one-source-out the ranking inverts: fabrication falls to 0.686 while omission holds at 0.793, and binary any-hallucination detection falls from 0.820 to 0.566, near chance. Reference-conditioned features are the only ones that recover binary detection (0.702), though they underperform on omission. Because we release per-judge verdicts and independent human labels on a shared subset, we can further show that the panel’s scene-fabrication label tracks report length (AUC 0.727) where the human label does not (0.566), and that assigning each source LLM a distinct judge pair, an unavoidable consequence of no-self-evaluation, confounds source LLM with measurement instrument. Axis-level signatures survive that confound; one fine-grained ordering does not. We release the corpus, per-judge verdicts, rubrics, reliability tiers, splits, and baselines.

1 INTRODUCTION

Large language models are increasingly evaluated by other large language models. Panel-of-judges protocols now label datasets far larger than any human annotation effort could reach, and the resulting labels underpin leaderboards, safety audits, and deployment decisions. Whether those labels measure what they are taken to measure is rarely examined, and the raw judge verdicts are rarely released for anyone to check.

Hallucination detection inherits both problems. A detector is trained on text labelled this way and tested on text from the same LLMs that produced the training set, split at random. Model families differ sharply in surface style and equally sharply in how often they hallucinate. A detector with access to both regularities can score well by recognising which family wrote a passage and applying that family’s base rate, without assessing whether any particular claim is supported. Under a random split this is invisible, because the models at test time are the models at training time. The reported accuracy may therefore describe how distinguishable the writers are rather than how detectable hallucination is.

The distinction matters wherever a detector is deployed against text from a model it has not seen, which is the normal case. It matters most in settings where the cost of an undetected error is borne by someone other than the system’s operator. Forensic reporting is one such setting: the difference between inventing a detail and omitting a crime is the difference between two kinds of evidentiary failure, and a screening tool that silently stops working when the underlying model is updated offers false assurance rather than none.

Separating the two possibilities requires evidence that existing resources cannot supply. It requires text from several model families over identical inputs, labelled per instance, so that the writer can be withheld while everything else is held fixed. Benchmarks built from a single model, or from several pooled without controlling the split, cannot distinguish a detector that reads evidence from one that reads style. We therefore ask:

1. Do hallucination detectors transfer to source LLMs they were not trained on, and by how much does performance change?
2. How does detectability depend on hallucination type, and does the type ordering hold under transfer?
3. Which surface properties carry the detectable signal, and what remains once source identity is removed?
4. How much of the loss does conditioning on the reference annotation recover, and for which types?
5. How does panel design shape the labels themselves, both in what drives judge verdicts and in what excluding self-evaluation implies?

Throughout, we refer to the LLM that produced a report as its *source LLM*, to distinguish it from the LLMs that later judge it. In the language of shortcut learning (Geirhos et al., 2020), source identity is a spurious feature: predictive in distribution, cheaply recoverable from surface style, and worthless once the source changes. Whether detectors rely on it is an empirical question, and the remainder of this paper answers it.

2 RELATED WORK

Existing hallucination benchmarks target short outputs and a fixed set of writers. CHAIR (Rohrbach et al., 2018) measures object hallucination in captions, POPE (Li et al., 2023b) uses polling, HaluEval (Li et al., 2023a) benchmarks 35K samples, THRONE (Kaul et al., 2024) unifies evaluation across six models, Bingo (Cui et al., 2023) identifies bias triggers, and VideoHalluciner (Wang et al., 2024) extends the setting to short clips. None varies the writing model while holding the input fixed, so none can separate a detector that reads evidence from one that reads style. Detection methods inherit the same limitation: SelfCheckGPT (Manakul et al., 2023), semantic entropy (Kuhn et al., 2023), and FActScore (Min et al., 2023) detect hallucination via consistency, uncertainty, or fact decomposition, but are evaluated within a single writer or across several pooled without a held-out condition. Our reference-conditioned baseline is closest to FActScore, simplified to lexical grounding.

Panel designs (Zheng et al., 2023) and cross-model panels that mitigate self-preference (Panickssery et al., 2024) are now standard, and their reliability is typically summarised by a single agreement figure. We report it per hallucination type against independent human raters, and show that the no-self-evaluation constraint itself induces a source-by-instrument confound that single-figure reporting hides. The underlying material comes from UCF-Crime (Sultani et al., 2018), which provides the video benchmark, and UCA (Yuan et al., 2024), whose fine-grained natural-language annotations serve as the reference; the reports and per-judge verdicts were generated by prior work (Anonymous, 2026) to measure prevalence, a study that considers neither detection, transfer, nor the reliability of its own labels. Section 3.1 states what is inherited and what is new.

What this paper adds. A corpus fully crossed over source LLMs and prompting strategies, released with per-judge verdicts; splits that withhold an entire source LLM; reference detectors spanning surface, lexical, embedding and reference-conditioned features; and a diagnosis of the labels themselves.

3 METHODOLOGY

3.1 TASK AND CORPUS

An instance is a triple (v, m, τ) : a surveillance video v , a source LLM m , and a prompting strategy τ . The source LLM produces a free-form forensic report r describing what occurs in v , and each

Table 1: Corpus composition. Every source LLM describes all 807 videos under all eight strategies, so the corpus is fully crossed. Gemini is seven reports short because of unrecoverable API failures.

Source LLM	Reports	Words	Mean words/report
Claude	6,456	6.40M	992
GPT	6,456	3.74M	580
Gemini	6,449	11.62M	1,802
Total	19,361	21.77M	1,124

report carries an expert reference annotation g_v written independently of any model. The task is to predict, from r and optionally g_v , six binary labels: H1 scene fabrication, H5 entity fabrication and H6 phantom actors, which invent content absent from the footage; H3 crime missed, which drops the criminal act; and H2 crime misclassification and H4 severity minimization, which misstate its nature or gravity. These group onto three axes, *fabrication*, *omission* and *distortion*. We report per-type AUC and F_1 , a macro over the six, a macro restricted to the reliable types (Section 3.2), and a binary any-hallucination score.

Provenance. The reports, taxonomy, panel protocol, and human-validation subset were introduced by prior work (Anonymous, 2026) to measure hallucination prevalence. HALT reuses them as raw material; new here are the detection task, the reliability tiering, the four splits, the baselines, and every analysis in Section 4.

The corpus draws 807 anomalous surveillance videos from UCF-Crime (Sultani et al., 2018) across eleven crime categories, paired with the fine-grained annotations of UCA (Yuan et al., 2024) as reference. Three frontier multimodal LLMs each analyse every video under eight prompting strategies (zero-shot, chain-of-thought, meta-prompting, self-consistency, sequential, ReAct, least-to-most, and iterative refinement), yielding 19,361 reports totalling 21.77M words. Two properties distinguish this from existing hallucination benchmarks: the outputs are long-form, averaging 1,124 words, and the corpus is fully crossed, so any comparison across source LLMs holds the input fixed by construction. Verbosity nonetheless varies by a factor of three under an identical task (Table 1), which is the regularity Section 4 shows detectors exploiting.

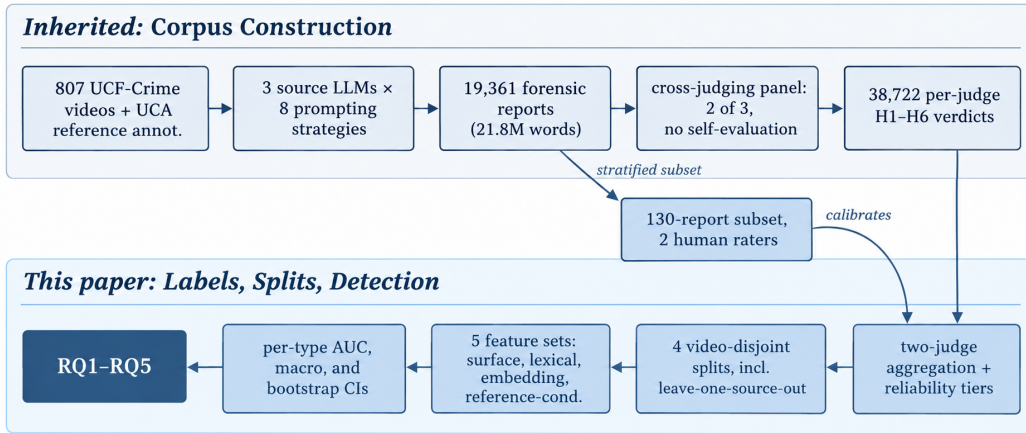


Figure 1: The HALT pipeline. The shaded band is inherited; the remainder is this paper: aggregating the raw verdicts into labels whose per-type reliability is calibrated against an independently scored subset, partitioning the corpus into four video-disjoint splits, and evaluating five feature sets on every split and every type.

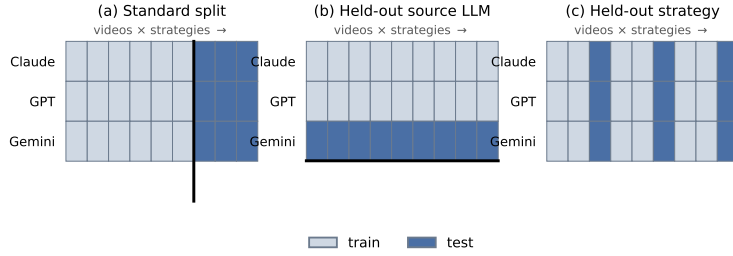


Figure 2: The corpus is fully crossed, so the same 19,361 reports admit several disjoint partitions. Each cell is one report; rows are source LLMs and columns are video-by-strategy conditions. (a) The standard split withholds labels, leaving every source LLM represented in training. (b) Withholding a source LLM asks whether a detector transfers to a model family it has never seen, which a standard split cannot test because the source LLMs at test time are the source LLMs at training time. (c) Withholding the multi-turn strategies tests transfer across prompting regimes. Existing hallucination benchmarks admit only (a).

3.2 LABELS AND THEIR RELIABILITY

Labels come from a panel of the same three models acting as judges (Zheng et al., 2023). Each report is scored by the two judges other than the model that produced it, so no model evaluates its own output and self-preference bias (Panickssery et al., 2024) is structurally excluded. This yields 38,722 verdicts, two per report, with zero self-judging rows, both verifiable from the released per-judge file. The released label is the two-judge majority with ties broken toward no hallucination, which with two judges is a conjunction; we release the per-judge verdicts as well, so alternative aggregation rules can be applied without rerunning any model. Judge inputs truncate the reference to 1,500 and the model output to 3,000 characters, and we release the pre-truncation length of every report so this can be conditioned on. A small number of cells carry a failure sentinel rather than a verdict; these are excluded from the vote and flagged.

Two human annotators independently scored a stratified subset of 130 reports spanning all source LLMs and strategies, working from the reference annotation and the report text with the panel’s labels hidden. For each type we compute agreement between the raters, and between the panel and the rater consensus, restricting the latter to reports where the raters agree. From these we assign each type a reliability tier and exclude the low tier from the headline macro. The tiers and what they imply for reading absolute rates are reported in Section 4.1 and released with the resource.

3.3 EVALUATION SETUP

Each video contributes 24 reports, so a split over reports would place the same video in training and test and let a detector memorise the video rather than learn the phenomenon. All splits are therefore video-disjoint (Figure 2). The **standard** split is crime-stratified 70/15/15 over videos. The **held-out source LLM** split trains on two source LLMs and tests on the third; we report all three folds, since a single fold confounds the difficulty of the held-out model with the transfer question. The **held-out strategy** split trains on the four single-turn strategies and tests on the four multi-turn ones. The **gold subset** of 130 human-annotated reports is flagged in the release as a small high-confidence evaluation set. The held-out-source condition is the diagnostic one: existing benchmarks draw from a single source LLM or pool several without controlling the split, so a detector can score well by recognising who wrote a report.

We provide five reference detectors, all logistic regression with balanced class weights, deliberately simple so that difficulty is attributable to the task rather than to under-tuned models. **Surface** uses verbosity, sentence statistics, hedging density, numeric detail density, type-token ratio, truncation length, and source and strategy identity indicators; **surface-noid** drops the identity features; **tfidf** uses word 1–2 gram TF-IDF; **embed** uses sentence embeddings; and **grounded** uses report-versus-reference features only, namely the rate of report tokens absent from the reference, the rate of reference tokens absent from the report, Jaccard overlap, reference coverage, length ratio, and

Table 2: Per-type label reliability on the 130-report human-annotated subset. κ_{panel} compares the panel against the human consensus and is computed only on reports where the two raters agree; n is that subset size. Tier thresholds are $\kappa \geq 0.65$ (high) and $\kappa \geq 0.40$ (medium). H1 is excluded from the headline macro score.

Type	Axis	n	$\kappa_{\text{human-human}}$	κ_{panel}	Tier
H1 Scene Fabrication	Fabrication	62	-0.008	0.150	low
H2 Crime Misclassification	Distortion	99	0.470	0.700	high
H3 Crime Missed	Omission	110	0.532	0.432	medium
H4 Severity Minimization	Distortion	103	0.406	0.583	medium
H5 Entity Fabrication	Fabrication	64	0.082	0.811	high
H6 Phantom Actors	Fabrication	84	0.192	0.658	high
Macro			0.279	0.556	

TF-IDF cosine. Identity features are dropped automatically on any split where they are constant in training and unseen at test.

We release HALT: the corpus with reference annotations, the per-judge verdicts, the aggregated labels, the judge and rater rubrics, the reliability tiers with supporting n , the splits, reference implementations of every baseline, the extended result tables cited in Section 4, and a datasheet (Geburu et al., 2021). The artifact is available at <https://anonymous.4open.science/r/HALT-ANON-7FF2/>.

4 RESULTS

4.1 CORPUS STATISTICS AND LABEL PROPERTIES

Under the released aggregation rule, 91.1% of reports carry at least one hallucination, with a mean of 2.58 per report. The rule matters: a disjunctive rule on the same verdicts gives 98.5% and 3.55 per report, a seven-point swing in the headline number that a release of aggregated labels alone would conceal. Output length differs by a factor of three across source LLMs under an identical task (Table 1), which is the regularity the rest of this section shows detectors exploiting.

Reliability is strongly type-dependent (Table 2): entity fabrication, crime misclassification, and phantom actors reach κ of 0.81, 0.70, and 0.66 against the human consensus, while scene fabrication reaches only 0.15 and is excluded from the headline macro. Two qualifications matter for anyone using the resource. The consensus κ is computed only on reports where the two raters agree, and that subset ranges from 62 to 110 of 130, so H5’s 0.81 rests on 64 reports. And the raters agree only modestly with each other on the fabrication types (macro $\kappa = 0.28$, with $\kappa \approx 0$ on H1), below the judges’ mutual agreement. The disagreement has a direction: because reference annotations are summary-level, a report describing genuinely observed but granular detail absent from the summary can be scored as fabrication, and on the consensus subset the panel flags scene fabrication in 62% of reports both humans call negative while almost never missing a human-confirmed one. Absolute fabrication prevalence should therefore be read as an upper bound; the relative quantities the benchmark supports are unaffected, since any labelling bias is shared across source LLMs.

Judge inputs cap model output at 3,000 characters and 65% of reports exceed it, with the most verbose source LLMs the most truncated. Truncation can therefore only understate their fabrication rates rather than manufacture them.

4.2 SAME-SOURCE DETECTION REPRODUCES PRIOR FINDINGS (RQ1)

Under the standard video-disjoint random split (released artifact, §B–C), fabrication is the most detectable axis. Surface features alone reach 0.858 on scene fabrication and 0.865 on entity fabrication, against 0.646 on crime omission and 0.653 on severity minimization. TF-IDF improves every type, reaching 0.846 macro and 0.820 binary. This ordering, fabrication detectable and omission resistant, matches what prior work reports and is what a practitioner would conclude from this corpus under conventional evaluation.

Table 3: Leave-one-source-out, TF-IDF features. The random-split row is repeated from the same-source baseline for comparison. Fabrication types (H1, H5) degrade four to five times as much as the omission and distortion types. Brackets give bootstrap 95% confidence intervals (1,000 resamples). The H3 interval does not overlap the H1 or H5 interval in any fold.

Held-out source LLM	H1	H2	H3	H4	H5	H6	ANY
Claude	0.596 [.581,.611]	0.718 [.704,.732]	0.787 [.776,.799]	0.788 [.774,.801]	0.671 [.658,.686]	0.778 [.765,.788]	0.539 [.502,.578]
GPT	0.702 [.689,.714]	0.735 [.719,.749]	0.796 [.785,.807]	0.779 [.767,.790]	0.719 [.707,.730]	0.736 [.713,.760]	0.548 [.531,.566]
Gemini	0.630 [.613,.644]	0.727 [.715,.741]	0.795 [.785,.806]	0.715 [.702,.729]	0.621 [.604,.640]	0.720 [.708,.732]	0.613 [.585,.636]
Mean (held-out)	0.643	0.727	0.793	0.761	0.671	0.744	0.566
Random split	0.882	0.773	0.854	0.822	0.888	0.859	0.820
Degradation	-0.239	-0.046	-0.061	-0.061	-0.217	-0.115	-0.254

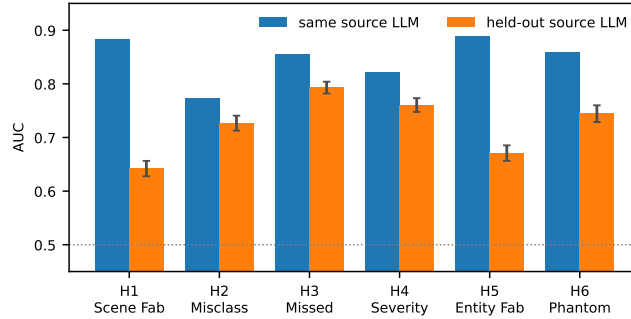


Figure 3: Per-type detector AUC (TF-IDF features) under the standard same-source split and under leave-one-source-out, with bootstrap 95% intervals on the held-out bars. The fabrication types (H1, H5) lose roughly four times as much as the omission and distortion types, inverting the ordering a same-source evaluation would report.

4.3 THE ORDERING INVERTS UNDER SOURCE-LLM SHIFT (RQ1–RQ3)

Leave-one-source-out reverses it (Table 3, Figure 3). Averaged over three folds, TF-IDF gives fabrication 0.686 and omission 0.793. Scene fabrication falls from 0.882 to 0.643 and entity fabrication from 0.888 to 0.671, while crime omission falls only from 0.854 to 0.793 and crime misclassification from 0.773 to 0.727. The types that looked easiest degrade four to five times as much as the types that looked hardest.

The separation is statistically unambiguous. Bootstrap 95% intervals (Table 3) place crime omission at [0.776, 0.799], [0.785, 0.807], and [0.785, 0.806] across the three folds, while scene fabrication never exceeds 0.714 and entity fabrication never exceeds 0.730. The omission interval overlaps neither fabrication interval in any fold, so each of the two principal fabrication types is significantly less detectable than omission under source-LLM shift.

The effect is not an artifact of residual video overlap. Because each video contributes 24 reports, holding out a source LLM leaves the same videos on both sides of the split. Re-running the three folds with video disjointness additionally enforced (training on 9,003 reports, testing on 1,016) lowers every number and widens the gap: fabrication falls to 0.668 and omission to 0.780, so the omission-minus-fabrication margin grows from 0.107 to 0.112, and binary detection falls to 0.559. The figures we report throughout are therefore the conservative ones.

Binary detection collapses outright. Any-hallucination AUC falls from 0.820 to 0.566, with per-fold values of 0.539, 0.548, and 0.613. On the held-out Claude fold the interval is [0.502, 0.578]: detecting whether a long-form report contains any hallucination at all, on a source LLM the detector has not seen, is statistically indistinguishable from a coin flip.

Table 4: Feature sets under leave-one-source-out, averaged over the three folds. Axis scores are the mean per-type AUC within the axis. No feature set wins on every axis; reference-conditioned features are the only ones that recover binary detection.

Features	Fabrication	Omission	Distortion	MACRO	ANY
surface	0.584	0.494	0.545	0.558	0.583
embed	0.669	0.719	0.658	0.674	0.595
tfidf	0.686	0.793	0.744	0.723	0.566
grounded	0.679	0.577	0.587	0.631	0.702
grounded+surface	0.691	0.607	0.599	0.646	0.652

Table 5: Does report length predict the panel label or the human label? AUC of a single feature, report word count, against each label source on the 130-report subset. The human column is restricted to reports where both raters agree (n from Table 2). A large positive gap means the feature tracks the judge rather than the phenomenon. Report length is the only feature with a positive gap of any magnitude; the reference-conditioned features diverge in the opposite direction; the remaining columns are in the released artifact, §D.

Type	AUC vs. panel	AUC vs. human	Gap
H1 Scene Fabrication	0.727	0.566	+0.161
H4 Severity Minimization	0.505	0.275	+0.230
H3 Crime Missed	0.484	0.390	+0.093
H2 Crime Misclassification	0.621	0.608	+0.013
H5 Entity Fabrication	0.784	0.800	−0.016
H6 Phantom Actors	0.662	0.668	−0.006

The mechanism is visible in the corpus statistics. Scene fabrication prevalence is 89.6% for Claude and 70.7% for Gemini against 30.9% for GPT, and mean report length varies from 580 to 1,802 words across the three. A detector that identifies the source LLM from style and applies its base rate achieves high apparent fabrication accuracy. Removing identity features does not remove the shortcut, since style is recoverable from the text itself (released artifact, §B–C).

Transfer also depends on whether a similar failure profile appears in training. Holding out GPT, the only omission-dominant family, drops embedding scene-fabrication AUC to 0.526 while raising crime-omission AUC to 0.824. Holding out Claude, leaving the similarly fabrication-dominant Gemini, raises scene fabrication to 0.853. Detectors generalise to a new source LLM mainly when the training pool contains one that fails the same way, which is an argument for benchmarks that span multiple source-LLM families rather than scaling a single one.

4.4 GROUNDING RECOVERS BINARY DETECTION BUT NOT OMISSION (RQ4)

Reference-conditioned features are the only ones that move binary detection under source-LLM shift, reaching 0.702 against 0.566 for TF-IDF, 0.595 for embeddings, and 0.583 for surface (Table 4). They also give the best fabrication axis, 0.691 when combined with surface features.

They are, however, worse on omission: 0.577 against TF-IDF’s 0.793. We expected the reference-token-coverage feature to capture crime omission directly and it does not. Whether a report drops the criminal event is apparently expressed in how the report is written rather than in bag-of-words coverage of the reference. No feature set wins everywhere, and the best macro under source-LLM shift is TF-IDF at 0.723, leaving substantial headroom for a detector that grounds claims in the footage rather than in a lexical reference. Binary detection under shift is thus the benchmark’s open problem, with omission and distortion under grounding a second target.

4.5 PANEL LABELS CARRY A LENGTH BIAS (RQ5)

Because we release panel labels and independent human labels on a shared subset of 130 reports, a single feature can be scored against both label sources (Table 5). Report length predicts the panel’s scene-fabrication label at AUC 0.727 but the human label at only 0.566, a gap of 0.161. Severity

minimization diverges more sharply still: for humans, length runs inverse to the label (0.275, well below chance), while the panel sees no relationship at all (0.505).

Entity fabrication and phantom actors show no such gap, 0.784 versus 0.800 and 0.662 versus 0.668. Panel and humans agree there that longer reports fabricate more. The bias is therefore specific rather than pervasive, which is what makes it diagnostic rather than merely deflationary: it identifies which labels to distrust rather than casting doubt on all of them.

The divergence is specific to verbosity rather than to disagreement in general. Repeating the comparison for the reference-conditioned features gives negative gaps on almost every type, reaching -0.236 on scene fabrication. On grounded quantities the panel does not diverge from the human raters at all; only report length predicts the judge better than it predicts a human reader.

The direction matches the labelling mechanism identified in Section 4.1: longer reports contain more granular detail absent from a summary-level reference, and that detail is scored as fabrication.

4.6 NO SELF-EVALUATION IMPLIES A SOURCE-BY-INSTRUMENT CONFOUND (RQ5)

Excluding self-evaluation has an unavoidable consequence: each source LLM is scored by a different pair of judges, and those pairs differ in mutual agreement by a factor of five on the fabrication types, from Cohen’s $\kappa = 0.850$ down to 0.148. Under a conjunctive aggregation rule a disagreeing pair produces fewer positives for purely mechanical reasons, so aggregated per-source rates confound the source LLM with its measurement instrument. To our knowledge this has not been reported, and it applies to any no-self-evaluation panel.

The confound is testable here because each judge scores exactly two source LLMs, permitting a within-judge comparison that holds the instrument fixed (released artifact, §B–C). Across 18 judge-by-type-by-pair comparisons, the source LLM ordering given by the aggregated labels is reproduced in 16. At the axis level it is reproduced in all six: every judge that sees GPT calls it omission-dominant, and every judge that sees Claude or Gemini calls them fabrication-dominant.

The two failures are informative. Both fall on the GPT judge and both on fabrication types, where GPT-as-judge flags nearly everything it sees (92.0% and 96.1% on scene fabrication for the two source LLMs it scores) and therefore cannot discriminate between them. The Claude-versus-Gemini fabrication ordering rests on a 4 to 7 point within-judge margin and is judge-dependent. The Claude-versus-GPT and Gemini-versus-GPT orderings, with within-judge gaps of 29 to 41 points reproduced by every judge, are robust. None of this is checkable without the raw verdicts, which cost nothing to release and foreclose nothing: the same verdicts yield 91.1% or 98.5% prevalence depending on the aggregation rule alone.

5 LIMITATIONS

Single corpus, three families. All reports derive from one video benchmark and one annotation set, and leave-one-source-out has three folds. The effect is consistent across all three, but three is a small sample for a claim about source-LLM families in general, and whether the inversion reproduces on other long-form tasks is untested.

Small human subset. The label-divergence analysis rests on 130 reports, with 62 to 110 per type after restricting to rows where both raters agree; gaps below roughly 0.10 should be treated as noise, and the two we report exceed it. Reference annotations are summary-level, which is the source of the fabrication labelling bias we quantify rather than eliminate.

Simple baselines. All detectors are linear; the comparison of interest is between splits, using the same detector on both sides.

6 CONCLUSION

Hallucination detectors are evaluated almost exclusively on text from the source LLMs they were trained on. HALT makes the alternative testable, and much of the reported skill turns out to be recognition of the writer rather than assessment of the claim: withholding a source LLM inverts the

per-type ordering and drives binary detection to the edge of chance. Grounding against the reference moves it only part of the way, and the labels carry a related weakness, tracking report length where human annotators do not. Held-out-source evaluation should be routine for any detector claimed to detect a phenomenon rather than a model, and per-judge verdicts released as standard.

ETHICS STATEMENT

The corpus derives from a publicly available surveillance benchmark of de-identified footage. Annotation was performed by members of the research group working from text. We release the resource for auditing and detector development. It is not suitable for, and we do not endorse, automated legal or investigative decision-making: a system whose error rate depends on the source LLM, the prompting strategy, and the event type does not meet the standard any evidentiary use would require, which is part of what these results establish.

REPRODUCIBILITY STATEMENT

A single entry point rebuilds every label from the released per-judge verdicts and regenerates every table and figure, verifying the published corpus statistics on each run so that divergence is caught immediately. All splits are seeded. Section 3.3 lists the released artifacts and gives the anonymous repository link.

REFERENCES

- Anonymous. Beyond accuracy: A type-aware taxonomy and evaluation framework for prompted hallucination in forensic multimodal llms. In *IEEE International Conference on Data Mining (ICDM)*, 2026. To appear; authors withheld for double-blind review.
- Chenhang Cui, Yiyang Zhou, Xinyu Yang, Shirley Wu, Linjun Zhang, James Zou, and Huaxiu Yao. Holistic analysis of hallucination in gpt-4v(ision): Bias and interference challenges. *arXiv preprint arXiv:2311.03287*, 2023.
- Timnit Gebru, Jamie Morgenstern, Briana Vecchione, Jennifer Wortman Vaughan, Hanna Wallach, Hal Daumé III, and Kate Crawford. Datasheets for datasets. *Communications of the ACM*, 64(12):86–92, 2021.
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. Shortcut learning in deep neural networks. *Nature Machine Intelligence*, 2(11):665–673, 2020.
- Prannay Kaul, Zhizhong Li, Hao Yang, Yonatan Dukler, Ashwin Swaminathan, C. J. Taylor, and Stefano Soatto. Throne: An object-based hallucination benchmark for the free-form generations of large vision-language models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
- Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations (ICLR)*, 2023.
- Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. Halueval: A large-scale hallucination evaluation benchmark for large language models. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023a.
- Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large vision-language models. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023b.
- Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

- Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. Factscore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- Arjun Panickssery, Samuel R. Bowman, and Shi Feng. Llm evaluators recognize and favor their own generations. *arXiv preprint arXiv:2404.13076*, 2024.
- Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. Object hallucination in image captioning. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2018.
- Waqas Sultani, Chen Chen, and Mubarak Shah. Real-world anomaly detection in surveillance videos. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 6479–6488, 2018.
- Yuxuan Wang, Yueqian Wang, Dongyan Zhao, Cihang Xie, and Zilong Zheng. Videohalluciner: Evaluating intrinsic and extrinsic hallucinations in large video-language models. *arXiv preprint arXiv:2406.16338*, 2024.
- Tongtong Yuan, Xuange Zhang, Kun Liu, Bo Liu, Chen Chen, Jian Jin, and Zhenzhen Jiao. Towards surveillance video-and-language understanding: New dataset, baselines, and challenges. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 22052–22061, 2024.
- Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging llm-as-a-judge with mt-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.